

```
from google.colab import files
```

```
uploaded = files.upload()
```

[Choose Files](#) StudentPerf...Factors.csv

StudentPerformanceFactors.csv(text/csv) - 641949 bytes, last modified: 9/30/2026 - 100% done
Saving StudentPerformanceFactors.csv to StudentPerformanceFactors.csv

```
import pandas as pd
```

Q1 Import a CSV dataset using Pandas.

```
data = pd.read_csv("StudentPerformanceFactors.csv")
```

```
print(data.head())
```

```

Hours_Studied  Attendance  Parental_Involvement  Access_to_Resources  \
0             23           84                   Low                High
1             19           64                   Low                Medium
2             24           98                  Medium                Medium
3             29           89                   Low                Medium
4             19           92                  Medium                Medium

Extracurricular_Activities  Sleep_Hours  Previous_Scores  Motivation_Level  \
0                        No             7                73                Low
1                        No             8                59                Low
2                       Yes             7                91                Medium
3                       Yes             8                98                Medium
4                       Yes             6                65                Medium

Internet_Access  Tutoring_Sessions  Family_Income  Teacher_Quality  \
0             Yes                  0             Low                Medium
1             Yes                  2             Medium              Medium
2             Yes                  2             Medium              Medium
3             Yes                  1             Medium              Medium
4             Yes                  3             Medium              High

School_Type  Peer_Influence  Physical_Activity  Learning_Disabilities  \
0     Public      Positive              3                No
1     Public      Negative              4                No
2     Public      Neutral               4                No
3     Public      Negative              4                No
4     Public      Neutral               4                No

Parental_Education_Level  Distance_from_Home  Gender  Exam_Score
0             High School              Near    Male         67
1              College             Moderate  Female         61
2             Postgraduate              Near    Male         74
3             High School             Moderate  Male         71
4              College              Near    Female         70

```

Q2.Find missing values in the dataset.

```
print(data.isnull().sum())
```

```

Hours_Studied      0
Attendance          0
Parental_Involvement  0
Access_to_Resources  0
Extracurricular_Activities  0
Sleep_Hours        0
Previous_Scores     0
Motivation_Level    0
Internet_Access     0
Tutoring_Sessions   0
Family_Income       0
Teacher_Quality     78
School_Type         0
Peer_Influence      0
Physical_Activity    0
Learning_Disabilities  0
Parental_Education_Level  90
Distance_from_Home  67
Gender              0
Exam_Score          0
dtype: int64

```

Q3.Replace missing numerical values with the mean.

```
data.fillna(data.mean(numeric_only=True), inplace=True)
print(data)
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	
...	...	...	...	...	
6602	25	69	High	Medium	
6603	23	76	High	Medium	
6604	20	90	Medium	Low	
6605	10	86	High	High	
6606	15	67	Medium	Low	

	Extracurricular_Activities	Sleep_Hours	Previous_Scores	\
0	No	7	73	
1	No	8	59	
2	Yes	7	91	
3	Yes	8	98	
4	Yes	6	65	
...	...	...	...	
6602	No	7	76	
6603	No	8	81	
6604	Yes	6	65	
6605	Yes	6	91	
6606	Yes	9	94	

	Motivation_Level	Internet_Access	Tutoring_Sessions	Family_Income	\
0	Low	Yes	0	Low	
1	Low	Yes	2	Medium	
2	Medium	Yes	2	Medium	
3	Medium	Yes	1	Medium	
4	Medium	Yes	3	Medium	
...	...	...	...	...	
6602	Medium	Yes	1	High	
6603	Medium	Yes	3	Low	
6604	Low	Yes	3	Low	
6605	High	Yes	2	Low	
6606	Medium	Yes	0	Medium	

	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	\
0	Medium	Public	Positive	3	
1	Medium	Public	Negative	4	
2	Medium	Public	Neutral	4	
3	Medium	Public	Negative	4	
4	High	Public	Neutral	4	
...	...	...	...	...	
6602	Medium	Public	Positive	2	
6603	High	Public	Positive	2	
6604	Medium	Public	Negative	2	
6605	Medium	Private	Positive	3	
6606	Medium	Public	Positive	4	

	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	\
0	No	High School	Near	
1	No	College	Moderate	
2	No	Postgraduate	Near	
3	No	High School	Moderate	
4	No	College	Near	

Q4.Remove duplicate records.

```
data.drop_duplicates(inplace=True)
print(data)
```

```

...
6602 Medium Yes 1 High
6603 Medium Yes 3 Low
6604 Low Yes 3 Low
6605 High Yes 2 Low
6606 Medium Yes 0 Medium

Teacher_Quality School_Type Peer_Influence Physical_Activity \
0 Medium Public Positive 3
1 Medium Public Negative 4
2 Medium Public Neutral 4
3 Medium Public Negative 4
4 High Public Neutral 4
...
6602 Medium Public Positive 2
6603 High Public Positive 2
6604 Medium Public Negative 2
6605 Medium Private Positive 3
6606 Medium Public Positive 4

Learning_Disabilities Parental_Education_Level Distance_from_Home \
0 No High School Near
1 No College Moderate
2 No Postgraduate Near
3 No High School Moderate
4 No College Near
...
6602 No High School Near
6603 No High School Near
6604 No Postgraduate Near
6605 No High School Far
6606 No Postgraduate Near

Gender Exam_Score
0 Male 67
1 Female 61
2 Male 74
3 Male 71
4 Female 70
...
6602 Female 68
6603 Female 69
6604 Female 68
6605 Female 68
6606 Male 64

```

Q5.Display the cleaned dataset.

```

print("Cleaned Dataset:")
print(data)

```

Cleaned Dataset:

```

Hours_Studied Attendance Parental_Involvement Access_to_Resources \
0 23 84 Low High
1 19 64 Low Medium
2 24 98 Medium Medium
3 29 89 Low Medium
4 19 92 Medium Medium
...
6602 25 69 High Medium
6603 23 76 High Medium
6604 20 90 Medium Low
6605 10 86 High High
6606 15 67 Medium Low

Extracurricular_Activities Sleep_Hours Previous_Scores \
0 No 7 73
1 No 8 59
2 Yes 7 91
3 Yes 8 98
4 Yes 6 65
...
6602 No 7 76
6603 No 8 81
6604 Yes 6 65
6605 Yes 6 91
6606 Yes 9 94

Motivation_Level Internet_Access Tutoring_Sessions Family_Income \
0 Low Yes 0 Low
1 Low Yes 2 Medium
2 Medium Yes 2 Medium

```

3	Medium	Yes	1	Medium
4	Medium	Yes	3	Medium
...	...	...	...	...
6602	Medium	Yes	1	High
6603	Medium	Yes	3	Low
6604	Low	Yes	3	Low
6605	High	Yes	2	Low
6606	Medium	Yes	0	Medium

	Teacher_Quality	School_Type	Peer_Influence	Physical_Activity	\
0	Medium	Public	Positive	3	
1	Medium	Public	Negative	4	
2	Medium	Public	Neutral	4	
3	Medium	Public	Negative	4	
4	High	Public	Neutral	4	
...	...	...	...	...	
6602	Medium	Public	Positive	2	
6603	High	Public	Positive	2	
6604	Medium	Public	Negative	2	
6605	Medium	Private	Positive	3	
6606	Medium	Public	Positive	4	

	Learning_Disabilities	Parental_Education_Level	Distance_from_Home	\
0	No	High School	Near	
1	No	College	Moderate	
2	No	Postgraduate	Near	
3	No	High School	Moderate	